

Education

University of California, Riverside

June 2019

- B.S. in Computer Science | GPA: 3.4 / 4.0
- Relevant Courses: Computer Graphics, Virtual Reality, Software Engineering, Operating Systems, Linear Algebra, Multivariable Calculus, Physics, Data Structures and Algorithms, Machine Learning

Work Experience

Visiting Researcher: Real-time Rendering for Virtual Humans

University of Southern California Institute for Creative Technologies. Playa Vista, CA June - Aug. 2018

- Developed the blend-shape (animation) pipeline in C# to view and animate facial rigs for the Vision and Graphics Lab's internal Unity-based rendering software
- Coded a graphical user interface using python scripts providing accessibility to animate facial scans
- Parsed through data files efficiently in C++ containing various facial expressions then inputted the collected information into the animation pipeline
- Updated the pipeline's developments through a git repository and communicated status reports, progress, and issues through daily logs

Computer Graphics Assistant. UC Riverside. Riverside, CA

Jan. - Feb. 2018

- Programmed POV-Ray scripts to render OpenGL and PhysBAM simulation examples containing MPM particles, rigid and deformable bodies created by Computer Graphics doctorate students and professors for their technical paper submissions
- Collected and compiled photo stills using Linux/Unix commands to produce movie files
- Organized a Kanban board to communicate status of renderings

Technical Projects

Pómo. Independent Study: Technical Director

Jan. 2018 - Present

- Create storyline for an original 3D animated short
- Illustrate concept art for the character, various assets, and environment
- Write Maya and Python scripts for character rigging and animation

Carnival Games. Virtual Reality Game Developer

Sept. - Dec. 2018

- Collaborated with a team of 3 to assemble the game logic in C# and Unity for Oculus Rift
- Designed 3D Ball Toss game assets in Maya and Unity
- Maximized an immersive gaming experience by incorporating sound effects and visual stimulation

Parking App. Android Developer

Sept. - Dec. 2018

- Built an Android app in Java that assists students find available parking at UC Riverside using the Android Studio development environment
- Generated parking trends from university's servers using data stored in Google's Firebase
- Directed a team of 4 in user interface design choices as scrum master

Skills

- C++, C#, Python, Git, OpenGL, Unity 3D, Matlab, Maya, Photoshop, LaTeX, Agile Methodologies, Scrum, Mac OSX, Windows, Linux, Computer Graphics, Virtual Reality

Associations

- Association for Computing Machinery, Member 2015 - Present
- Society of Hispanic Professional Engineers, Member 2016 - Present
- Intramural Soccer, Captain and Member 2016 - Present